LAB 3 – ARCHITECTURAL PATTERN SELECTION & COMPONENT MODELLING

Community Tool & Equipment Library

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1. Architecture Selection

Chose Layered Architecture for the Community Tool & Equipment Library system.

2. Architectural Choice and Reasons

First, the system has a clear separation between user interaction, library operations, and stored information. A presentation layer can handle the Community Member and Library Custodian interactions, while the business layer handles reservations, deposits, returns, and damage assessment.

Second, the system is relatively small and has a centralized set of library data such as tool availability, loan dates, deposits, and damage records. Layering keeps these responsibilities organized and makes the system easier to understand, maintain, and test than introducing the operational complexity of microservices.

3. Security Advantage

The layered structure allows access to the database only through controlled business components. For example, members do not directly access inventory or deposit records. The business layer can validate member actions, loan limits, deposit calculations, and overdue status before data is stored or retrieved.

4. Performance Benefit

Frequently used operations such as checking tool availability and retrieving pricing or deposit information can be handled through the business and data layers without unnecessary distributed network calls. This reduces communication overhead and provides predictable response time for common library operations.

5. Main Components and Interfaces

The diagram contains six components: Member/Custodian Interface, Loan & Reservation Manager, Inventory Manager, Deposit & Fee Service, Return & Damage Manager, and Library Database, with an Audit/Notification Store for recorded events. Interfaces show user requests, loan/reservation processing, inventory availability, deposit/payment processing, and data updates. UML provided interfaces are shown with a lollipop and required interfaces with a socket.

6. Conclusion

Layered Architecture is suitable because the application has straightforward business rules and centralized data. It provides separation of concerns, controlled data access, and sufficient performance while keeping the design simple for a community library system.